Lecture 12

Morita equivalence and the Drinfeld centre

Objectives. *A fusion category C acting on a module category $\mathcal{M}$ (Lecture 11) has a second symmetry category, its* dual*, whose objects are the module endofunctors of $\mathcal{M}$. For* Vec_G *acting on* $\mathrm{Vect}_{\mathrm{fd}}$ *the dual is* $\mathrm{Rep}(G)$*. These categories have equivalent* Drinfeld centres*. An object of the centre $\mathcal{Z}(C)$ is an object of C equipped with a compatible family of crossings with every other object. The chosen crossings are additional data. The centre is braided, and modular when C is spherical. For* Vec_G *with $G=\mathbb{Z}_2$, its anyons are those of the toric code of* Quantum Computing*.*

By the end of this lecture you will be able to:

- Define the dual category of a module category, and identify $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}}$ with $\mathrm{Rep}(G)$.
- Define half-braidings and the Drinfeld centre, and prove that it is braided.
- Compute $\mathcal{Z}(\mathrm{Vec}_G)$ for $G=\mathbb{Z}_2$ by half-braidings, and identify it with the toric code, braiding and spins included.
- State Müger's theorem on the centre and the theorem of Etingof, Nikshych and Ostrik on Morita equivalence, and use them on Vec_G, $\mathrm{Rep}(G)$ and Fibonacci.

12.1 Module functors; the dual category; Morita equivalence

Module functors $(F,s)\colon \mathcal{M}\to\mathcal{N}$ were defined in Lecture 11 (§ 11.2). A *module natural transformation* $\xi\colon F\to F'$ is a natural transformation with

$$s'_{X,M}\circ \xi_{X\otimes M} = (\mathrm{id}_X\otimes \xi_M)\circ s_{X,M}.$$

Definition 12.1 (Dual category)**.** For a C-module category $\mathcal{M}$, the *dual category* $C^*_{\mathcal{M}} = \mathrm{Fun}_C(\mathcal{M},\mathcal{M})$ has the module endofunctors as objects and the module natural transformations as morphisms, with tensor product the composition, $F\otimes G = F\circ G$ (EGNO Def. 7.12.2).

Theorem 12.2 (Two duals · *cited; (ii) proved*)**.** *(i) For the regular module, $X\mapsto(-\otimes X)$ is a tensor equivalence $C^{\otimes\mathrm{op}}\simeq C^*_C$, where $C^{\otimes\mathrm{op}}$ has the reversed tensor product (EGNO Ex. 7.12.3). For* Vec_G*, $g\mapsto(-\otimes\delta_{g^{-1}})$ identifies* Vec_G *with* $(\mathrm{Vec}_G)^*_{\mathrm{Vec}_G}$*. (ii) For* Vec_G *acting on* $\mathrm{Vect}_{\mathrm{fd}}$ *by $\delta_g\otimes V=V$,* $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}}\simeq\mathrm{Rep}(G)$ *(EGNO Ex. 7.12.19).*

Proof of (ii), and the order in (i). In (i), $(-\otimes\delta_{g^{-1}})\circ(-\otimes\delta_{h^{-1}}) = -\otimes\delta_{h^{-1}g^{-1}} = -\otimes\delta_{(gh)^{-1}}$, so the assignment respects the tensor product, while $g\mapsto(-\otimes\delta_g)$ reverses it for non-abelian G; that it is an equivalence is EGNO's.

For the second dual, put $V = F(\mathbb{C})$. For $w \in W$, let $f_w \colon \mathbb{C} \to W$ send 1 to w. The natural map $V \otimes W \to F(W)$ is given by

$$v \otimes w \longmapsto F(f_w)(v).$$

It is an isomorphism because F is linear and preserves finite direct sums. The module structure is determined by the maps $s_g \colon F(\delta_g \otimes \mathbb{C}) \to \delta_g \otimes F(\mathbb{C})$, hence by $s_g \in \mathrm{GL}(V)$. With trivial module associators, compatibility (EGNO (7.6)) reads $s_{gh} = s_h\, s_g$. Thus $\rho(g) = s_g^{-1}$ is a representation.

Conversely, a representation (V, ρ) gives the functor $F_V(W) = V \otimes W$, with $F_V(f) = \mathrm{id}_V \otimes f$ and $s_g = \rho(g)^{-1}$ on V. Extend the module structure to graded direct sums using the ordinary flip of vector spaces. The representation law gives the required compatibility. These constructions are inverse up to the natural isomorphism above. A module natural transformation is determined by an intertwiner on V. Composing F_V with F_U gives $V \otimes U \otimes W$, with representation $\rho_V \otimes \rho_U$. ∎

So one symmetry, Vec_G, has two duals: itself (from the regular module) and $\mathrm{Rep}(G)$ (from $\mathrm{Vect}_{\mathrm{fd}}$).

Example 12.3 (The duals of $\mathrm{Vec}_{\mathbb{Z}_2}$). Lecture 11 found two indecomposable module categories over $\mathrm{Vec}_{\mathbb{Z}_2}$: the regular one and $\mathrm{Vect}_{\mathrm{fd}}$. Their duals are $\mathrm{Vec}_{\mathbb{Z}_2}$ (the group is abelian, so $\otimes$op changes nothing) and $\mathrm{Rep}(\mathbb{Z}_2)$, whose two simple objects, the trivial and the sign representation, fuse like $\mathbb{Z}_2$ with the trivial associator of vector spaces: $\mathrm{Rep}(\mathbb{Z}_2) \simeq \mathrm{Vec}_{\mathbb{Z}_2}$ as fusion categories ($\omega = 1$, Lecture 3). So $\mathrm{Vec}_{\mathbb{Z}_2}$ is dual to itself in two different ways. In the toric code of § 12.3 this is the exchange of e and m, and in Lecture 13 the Kramers–Wannier duality of the Ising chain.

To compare categories through module categories we need products of categories.

Definition 12.4 (Deligne product, bimodule category). For fusion categories $\mathcal{C}$ and $\mathcal{D}$, the *Deligne product* $\mathcal{C} \boxtimes \mathcal{D}$ is the fusion category with simple objects $a \boxtimes b$, for simple a in $\mathcal{C}$ and b in $\mathcal{D}$, and $(a \boxtimes b) \otimes (a' \boxtimes b') = (a \otimes a') \boxtimes (b \otimes b')$. Its morphism spaces are

$$\begin{aligned} &\mathrm{Hom}(X \boxtimes Y, X' \boxtimes Y') \\ &= \mathrm{Hom}_{\mathcal{C}}(X, X') \otimes \mathrm{Hom}_{\mathcal{D}}(Y, Y'). \end{aligned}$$

Composition, associators and unitors are taken componentwise, and objects are finite direct sums of the simple pairs (EGNO §1.11, §4.6). A $(\mathcal{C}, \mathcal{D})$-*bimodule category* is a module category over $\mathcal{C} \boxtimes \mathcal{D}^{\otimes\mathrm{op}}$ (EGNO Def. 7.1.7, Ex. 7.4.3).

Definition 12.5 (Morita equivalence). Fusion categories $\mathcal{C}$ and $\mathcal{D}$ are *Morita equivalent* if there is an indecomposable semisimple $\mathcal{C}$-module category $\mathcal{M}$ with $\mathcal{D}^{\otimes\mathrm{op}} \simeq \mathcal{C}^*_{\mathcal{M}}$ as tensor categories (EGNO Def. 7.12.17). Then $\mathcal{M}$ is a $(\mathcal{C}, \mathcal{D})$-bimodule category.

Without $\otimes$op the centres of § 12.4 would match only up to reversing the braiding. By Theorem 12.2, Vec_G and $\mathrm{Rep}(G)$ are Morita equivalent, since $\mathrm{Rep}(G)^{\otimes\mathrm{op}} \simeq \mathrm{Rep}(G)$ by the flip. Physically, Morita equivalent symmetry categories describe the same two-dimensional bulk, seen through different boundaries · *physical principle* (Kitaev and Kong; Lecture 13).

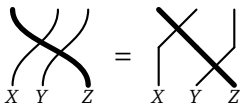
12.2 The Drinfeld centre

Definition 12.6 (Half-braiding, Drinfeld centre)**.** A *half-braiding* on an object Z of C is a natural isomorphism $\gamma_X : X \otimes Z \to Z \otimes X$ such that

$$\gamma_{X\otimes Y} = \alpha_{Z,X,Y} \circ (\gamma_X \otimes \mathrm{id}_Y) \circ \alpha^{-1}_{X,Z,Y} \circ (\mathrm{id}_X \otimes \gamma_Y) \circ \alpha_{X,Y,Z}$$

for all X, Y. The *Drinfeld centre* $\mathcal{Z}(C)$ has objects the pairs (Z, γ) and morphisms the $f : Z \to Z'$ with $(f \otimes \mathrm{id}_X) \circ \gamma_X = \gamma'_X \circ (\mathrm{id}_X \otimes f)$ (EGNO Def. 7.13.1).

A half-braiding is a strand Z, drawn thick, that every strand of C can cross. It is not a braiding of C: only Z carries one. We initially draw this crossing without over or under; the braiding of the centre below will use the over-crossing convention of Lecture 7. The condition says that crossing $X \otimes Y$ is crossing Y, then X:



(12.1)

The tensor product of (Z, γ) and (Z', γ') is $Z \otimes Z'$, with X crossing Z and then Z', and the unit is 1 with the trivial half-braiding (EGNO (7.42)).

Theorem 12.7 (The centre is braided · *proved given coherence*)**.** *$\mathcal{Z}(C)$ is a monoidal category, and*

$$c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z : Z \otimes Z' \to Z' \otimes Z$$

is a braiding on it (EGNO Prop. 8.5.1). In pictures, the strand Z crosses the thick strand Z', over it as in Lecture 7, by Z''s half-braiding:

$$c_{(Z,\gamma),(Z',\gamma')} \;=\; \;=\; \gamma'_Z. \qquad (12.2)$$

Proof. γ'_Z is a morphism of $\mathcal{Z}(C)$: for any X, pulling X across $Z \otimes Z'$ and then exchanging Z past Z' equals exchanging first, by naturality of γ' in its argument (applied to γ_X) and the condition (12.1) for γ'. It is natural in the first variable because γ' is natural in its argument, and in the second because morphisms of the centre commute with half-braidings. That the tensor product of two half-braidings is again one is the same kind of check, with the pentagon (EGNO (7.42)). The two hexagons of Definition 7.2: $c_{(Z,\gamma),(Z',\gamma')\otimes(Z'',\gamma'')}$ is $\gamma' \otimes \gamma''$ evaluated on Z, which by the definition of the tensor product is Z crossing Z' then Z''; and $c_{(Z,\gamma)\otimes(Z',\gamma'),(Z'',\gamma'')} = \gamma''_{Z\otimes Z'}$ is Z' then Z crossing Z'', which is (12.1) for γ''. In pictures both are strands sliding through crossings. ∎

The forgetful functor $\mathcal{Z}(C) \to C$ keeps Z and forgets γ. If C is already braided, $X \mapsto (X, c_{-,X})$ is a braided functor $C \to \mathcal{Z}(C)$ (EGNO Prop. 8.6.1). Two centres must not be confused: the Drinfeld centre $\mathcal{Z}(C)$ is a new category built from any monoidal C; the Müger centre of Lecture 7 is a subcategory of a braided one, its transparent objects.

12.3 $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ in full; $D(G)$ and $\mathcal{Z}(\mathrm{Vec}_G)$

Theorem 12.8 (The centre of Vec_A · *proved given coherence*). *For an abelian group A, the simple objects of $\mathcal{Z}(\mathrm{Vec}_A)$ are the pairs (g,χ), $g\in A$, χ a character of A, with $\gamma_{\delta_h}=\chi(h)\,\mathrm{id}$ on $\delta_h\otimes\delta_g$. Their braiding and spin are*

$$c_{(g,\chi),(g',\chi')}=\chi'(g),\qquad \theta_{(g,\chi)}=\chi(g).$$

For $A=\mathbb{Z}_2$, write the group additively as $\{0,1\}$: the generator s of Lecture 3 is now 1, and its identity is 0. The four simple objects $1=(0,1)$, $e=(0,\chi_-)$, $m=(1,1)$ and $\varepsilon=(1,\chi_-)$ have $\theta=1,1,1,-1$, $e\otimes m=\varepsilon$ and double braiding $c_{m,e}c_{e,m}=-1$.

Proof. Every morphism of Vec_A preserves the grading, so an object of the centre is a sum of pieces $\delta_g\otimes V$, and γ_{δ_h} maps $\delta_h\otimes(\delta_g\otimes V)$ to $(\delta_g\otimes V)\otimes\delta_h$. Since A is abelian both are $\delta_{hg}\otimes V$, so γ_{δ_h} is a map $\gamma_h\in\mathrm{GL}(V)$. The associators are trivial, so (12.1) reads $\gamma_{hk}=\gamma_h\gamma_k$: a representation of A, which splits into characters. So the simple objects are the (g,χ), and their tensor product multiplies both, $(g,\chi)\otimes(g',\chi')=(gg',\chi\chi')$; in particular $e\otimes m=(1,\chi_-)=\varepsilon$. By Theorem 12.7 $c_{(g,\chi),(g',\chi')}=\gamma'_{\delta_g}=\chi'(g)$, and for an invertible object the spin is its self-braiding, $\chi(g)$ (Theorem 7.7). For $\mathbb{Z}_2$: $\theta_\varepsilon=\chi_-(1)=-1$, $c_{e,m}=\chi_m(0)=1$ and $c_{m,e}=\chi_-(1)=-1$. ∎

These are the anyons of the toric code (Kong–Zhang (4.1.3), Ex. 4.2.17). In full, with rows X and columns Y in the order $1,e,m,\varepsilon$, the braiding $c_{X,Y}=\chi_Y(g_X)$ is

$$\left(c_{X,Y}\right)=\begin{pmatrix}1&1&1&1\\1&1&1&1\\1&-1&1&-1\\1&-1&1&-1\end{pmatrix}.$$

It is not symmetric: $c_{e,m}=1$ but $c_{m,e}=-1$. For $X\neq Y$, neither of them 1, a single entry can depend on the gauge, but the double braiding $c_{Y,X}c_{X,Y}$, the product of the matrix and its transpose entry by entry, does not, nor do the self-braidings on the diagonal; the double braiding is -1 for e and m, e and ε, m and ε. The double braiding -1 of e and m is the phase *Quantum Computing* measured by taking e around m, and the S-matrix of Example 10.14 follows from these numbers (Sheet 12). The anyons of the toric code are the centre of its symmetry $\mathbb{Z}_2$. Exchanging e and m preserves the fusion rules, the spins and the double braidings; it is a braided autoequivalence (Kong–Zhang §4.3.2; Sheet 12), the shadow of the two duals of Example 12.3. Twisting by $\omega(s,s,s)=-1$, on the generator s of $\mathbb{Z}_2$, changes the centre: the same computation, with associators, gives spins $1,1,i,-i$, the *double semion* (`scripts/check-l12-algebra.py`).

For a non-abelian G a simple δ_g need not carry a half-braiding at all: γ_{δ_h} must map $\delta_h\otimes\delta_g=\delta_{hg}$ to $\delta_g\otimes\delta_h=\delta_{gh}$, which is zero unless $hg=gh$. So δ_g alone works only for central g; in general the object must be a sum $\bigoplus_{g'\in[g]}\delta_{g'}\otimes V_{g'}$ over a whole conjugacy class, and the half-braiding is then fixed by a representation of the centraliser $C_G(g)$ on V_g. This is why the labels below are classes and centraliser representations. The centre is the representation category of a Hopf algebra.

Theorem 12.9 (The Drinfeld double · *cited*). *The* Drinfeld double $D(G)=\mathbb{C}^G\rtimes\mathbb{C}[G]$ *has basis $\delta_g\otimes h$ and product $(\delta_g\otimes h)(\delta_{g'}\otimes h')=\delta_{g,hg'h^{-1}}\,\delta_g\otimes hh'$; it is quasitriangular (EGNO Ex. 8.3.9). There are braided equivalences* $\mathrm{Rep}D(G)\simeq\mathcal{Z}(\mathrm{Vec}_G)\simeq\mathcal{Z}(\mathrm{Rep}G)$ *(EGNO Prop. 7.14.6, Ex. 8.5.4 and 8.5.6). The simple objects are the pairs $([g],\pi)$ of a conjugacy class and an irreducible representation of the centraliser $C_G(g)$ (*Symmetries *Defs. 2.4, 2.11), with $\theta_{([g],\pi)}=\chi_\pi(g)/\dim\pi$ (Bakalov–Kirillov (3.2.8)).*

The labels are not pairs (g, an irrep of G): for S_3 the classes of 1, the transpositions and the 3-cycles have centralisers S_3, $\mathbb{Z}_2$ and $\mathbb{Z}_3$, with $3+2+3=8$ simple objects (Sheet 12), and $\dim \mathcal{Z}(\mathrm{Vec}_G) = |G|^2$.

12.4 Müger and ENO (statements)

For a braided C, the *reverse* C^{rev} has the braiding $c^{\mathrm{rev}}_{X,Y} = c^{-1}_{Y,X}$ (EGNO Def. 8.1.4), and conjugate spins.

Theorem 12.10 (Müger · *cited*)**.** *Let C be a fusion category over $\mathbb{C}$. (i) $\mathcal{Z}(C)$ is non-degenerate, that is, its Müger centre is trivial (EGNO Cor. 8.20.13), and $\dim \mathcal{Z}(C) = (\dim C)^2$ (EGNO Prop. 9.3.4). (ii) If C is spherical, $\mathcal{Z}(C)$ is modular (EGNO Cor. 8.20.14). (iii) If C is modular, $\mathcal{Z}(C) \simeq C \boxtimes C^{\mathrm{rev}}$ as braided categories (Müger 2012, Thm. 7.1(vi)).*

"Modular" needs the spherical structure of Lecture 10. For Fibonacci, $\mathcal{Z}(\mathrm{Fib}) \simeq \mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ has dimension $(2+\varphi)^2$ and central charge $\frac{14}{5} - \frac{14}{5} = 0$ (the script checks the Gauss sum). The toric code, too, has $c = 0$: for spherical C the Gauss sums of $\mathcal{Z}(C)$ are $\dim C$, so $c = 0$ (Müger 2012, Thm. 7.1(iv)).

Theorem 12.11 (Etingof, Nikshych and Ostrik · *cited*)**.** *Fusion categories C and $\mathcal{D}$ are Morita equivalent if and only if $\mathcal{Z}(C) \simeq \mathcal{Z}(\mathcal{D})$ as braided categories (EGNO Thm. 8.12.3).*

So $\mathcal{Z}(\mathrm{Vec}_G) \simeq \mathcal{Z}(\mathrm{Rep}G)$ is Morita equivalence seen from the centre, and $\mathrm{Vec}_{\mathbb{Z}_2}$ and its twist by ω are *not* Morita equivalent: the double semion has spins $\pm i$, the toric code none. Physically, a gapped two-dimensional phase whose boundary carries the symmetry C has the anyons of $\mathcal{Z}(C)$ in its bulk, and every indecomposable module category is a gapped boundary of it · *physical principle* (Kitaev and Kong, §4; the bulk of a string net is the centre, Simon p. 485): the toric code is the bulk of $\mathrm{Vec}_{\mathbb{Z}_2}$, whose two module categories are its two boundaries, and Morita equivalent symmetries share a bulk. Lecture 13 makes this precise on the lattice.

Summary.

- *Handed on:* the dual $C^*_{\mathcal{M}} = \mathrm{Fun}_C(\mathcal{M},\mathcal{M})$, with $(\mathrm{Vec}_G)^*_{\mathrm{Vect_{fd}}} \simeq \mathrm{Rep}(G)$; Morita equivalence $\mathcal{D}^{\otimes\mathrm{op}} \simeq C^*_{\mathcal{M}}$ (Definitions 12.1 and 12.5 and Theorem 12.2).
- *The centre:* half-braidings $\gamma_X : X \otimes Z \to Z \otimes X$; $\mathcal{Z}(C)$ is braided by $c = \gamma'_Z$ (Definition 12.6 and Theorem 12.7).
- *The toric code:* $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ has $1, e, m, \varepsilon$ with $\theta = \chi(g)$ and double braiding -1 for e, m (Theorem 12.8); in general $\mathcal{Z}(\mathrm{Vec}_G) \simeq \mathrm{Rep}D(G)$, labelled by classes and centraliser irreps (Theorem 12.9).
- *Theorems:* the centre is non-degenerate, of dimension $(\dim C)^2$, modular if C is spherical, and $C \boxtimes C^{\mathrm{rev}}$ if C is modular; Morita equivalence is braided equivalence of centres (Theorems 12.10 and 12.11).
- *Next:* Lecture 13 reads module categories as boundaries and defects, and the dual category as the symmetry on them.

References

Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: the Deligne product, §1.11, p. 15, and §4.6, p. 73; bimodule categories, Def. 7.1.7, p. 133, and Ex. 7.4.3, p. 136; the dual category, §7.11, p. 154, Def. 7.12.2 and Ex. 7.12.3, p. 155; Morita equivalence, Def. 7.12.17, p. 157; $(\mathrm{Vec}_G)^*_{\mathrm{Vect}_{\mathrm{fd}}} \simeq \mathrm{Rep}(G)$, Ex. 7.12.19, p. 158; the centre, Def. 7.13.1, p. 162, and (7.42); Rep$D(H)$, Prop. 7.14.6, p. 165; the reverse braiding, Def. 8.1.4, pp. 195–196; $D(G)$, Ex. 8.3.9, p. 199; the braiding of the centre, Prop. 8.5.1, p. 207; $\mathcal{Z}(\mathrm{Vec}_G)$ and Rep$D(H)$, Ex. 8.5.4 and 8.5.6, pp. 207–208; braided $C \to \mathcal{Z}(C)$, Prop. 8.6.1, p. 208; Morita and centres, Thm. 8.12.3, p. 222; non-degenerate and modular centres, Cor. 8.20.13–8.20.14, p. 243; $\dim \mathcal{Z}(C)$, Prop. 9.3.4, p. 280. Kitaev and Kong, *Models for gapped boundaries and domain walls*, arXiv:1104.5047, §§4–5, pp. 8–17. Müger, *Modular categories* (2012): the centre, Def./Prop. 4.5, p. 9 (his half-braiding sits on the other side, so his centre is our $\mathcal{Z}(C)^{\mathrm{rev}}$), and Thm. 7.1, p. 21. Kong and Zhang, arXiv:2205.05565: the toric code from half-braidings, (4.1.3), p. 77 (their fermion is f); Ex. 4.2.17, p. 84; the $e \leftrightarrow m$ autoequivalence, §4.3.2, p. 91. Bakalov and Kirillov, *Lectures on Tensor Categories and Modular Functors*: $D(G)$, §3.2, pp. 59–61. Simon, *Topological Quantum*: quantum doubles, pp. 455–458; Morita equivalence and bulks, p. 485.